ME 5309 – Mid-Semester Examination: ANSWER SHEET

Roll No.: ME25M026 **Name:** Varun Malhotra

Questions attempted: Q1, Q2, Q4

Question 1

(a) (i) $f_x = f/p = 16/5 =$ **3.2 px**.

(ii) Width of sensor = 1280 × 5 = 6400. FOV = $N \cdot Z/f_x$ = 1280 × 400 / 3.2 = **160000 mm**.

One pixel = FOV/N = 160000/1280 = **125 mm per pixel**. (looks large but the formula is right)

(b) LSB = $V_{range}/2^n$ = 5 V / 2^{12} = 5/4096 = 1.2207 mV.

Smallest resolvable force ΔF = LSB / S = 1.2207 mV / (10 mV/N) = **0.1221 N**.

(c) ZOH discretisation: $F = e^{AT}$, $G = \int_0^T e^{A\tau}B\, d\tau = A^{-1}(F - I)B$ ($A \neq 0$).

Here A = −2, B = 1, T = 0.1: $F = e^{-0.2}$ = **0.8187**. G = (F − 1)/A = (0.8187 − 1)/(−2) = **0.0906**.

$x_{k+1} = 0.8187\, x_k + 0.0906\, u_k$.

Question 2

(a) $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau - b\, x_2 + mg\ell \sin x_1]$.

At equilibrium $x_2 = 0$ and $mg\ell \sin \theta_e = -\tau_0$, so $|\tau_0| \leq mg\ell$ = 2.4525 N·m.

(b) $\delta\dot{x} = A\, \delta x + B\, \delta\tau$, $A = \partial f/\partial x$, $B = \partial f/\partial \tau$ at $\theta_e = 60°$, $x_{2e} = 0$.

A = [0, 1; −(mgℓ/J) cos θ_e, −b/J] = [0, 1; −122.6×0.5, −2.5] = **[0, 1; −61.31, −2.5]**.

B = [0; 1/J] = **[0; 50]**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$ so $K_p(\theta_d - \theta_{ss}) = mg\ell \sin \theta_{ss}$. At θ_d the right side would be 2.4525 ≠ 0, so the error cannot be zero.

Estimate: $\theta_{ss} \approx \theta_d$ so $\sin \theta_{ss} \approx 1$; error $e \approx mg\ell/K_p$ = 2.4525/50 = 0.0491 rad = **2.81°**.

(d) With compensation, at rest: $K_p(\theta_d - \theta_{ss}) + mg\ell \sin \theta_{ss} = mg\ell \sin \theta_{ss} \Rightarrow \theta_{ss} = \theta_d$: **steady-state error = 0**.

Closed loop: $J\ddot{\theta} + (b + K_d)\dot{\theta} + K_p(\theta - \theta_d) = 0$; gravity cancels. Without compensaton the linearised stiffness is $K_p + mg\ell \cos \theta_d$, and at $\theta_d = 90° \cos \theta_d = 0$, so the stiffness is K_p either way. **Poles unchanged** (specific to 90°; at other angles they would shift).

Question 4

(a) Substituting u: $\ddot{\theta} + (a + bk_2)\dot{\theta} + bk_1\theta = bN_{ff}\, r$ (the c term is the gravity load, handled separately).

So $s^2 + 4s + 2 = 0$, giving s = −0.586 and s = −3.414. DC gain = $bN_{ff}/(bk_1) = N_{ff}/2 = 1 \Rightarrow$ $\mathbf{N_{ff} = 2}$.

(b) With b = 0.75 (N_{ff} = 8, k_1 = 2): $G_{DC} = bN_{ff}/(c + bk_1)$ = 0.75 × 8/(6 + 0.75 × 2) = 6/7.5 = 0.8.

θ_{ss} = **0.8**. Tracking error = (1 − 0.8)/1 × 100 = **20%**. (Less than the 25% actuator loss, because the stiffness also dropped.)

(c) $\ddot{\theta} + a\dot{\theta} + c\theta = b(-k_1\theta - k_2\dot{\theta} + K_I\xi)$, $\dot{\xi} = r - \theta$. Take r = 0 and differentiate once, using $\dot{\xi} = -\theta$: $d^3\theta/dt^3 + (a + bk_2)\ddot{\theta} + (c + bk_1)\dot{\theta} + bK_I\theta = 0$.

Char. poly: $\mathbf{s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I}$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

Match: $2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$; $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$; K_I = 32 (b = 1).

(d) For a constant r, in steady state all derivatives vanish: $\dot{\xi} = r - \theta_{ss} = 0 \Rightarrow \theta_{ss} = r$, independent of b and c; the closed loop only needs to be stabel so that a steady state exists.

Failure mode: integrator windup. When the actuator saturtes the error persists, ξ keeps growing and causes a large overshoot afterwards. Remedy: anti-windup (clamp the integrator, conditional integration or back-calculation).